

UI Design Progress Report 2 for Final Project

Professor Brian Smith

12/03/2021

Part 1: Updated Progress Report 1

1. Team Information

- Team Number: 26
- Project Name: “Media Map”
- Members: Je Seung You (jy2908), Alexsandra Gao (atg2152), Adi Gal (aeg2225), Swan Htet (sh3969)

2. Prior Individual Project

We are expanding on Adi’s individual project. We are reassessing the target user population by focusing on movies and TV shows exclusively (rather than books, movies, and TV shows), and we will expand on the project by doing more research with our users and by creating a basic implementation.

3. Target User Population

Our target user population is made of people in the teen to older adult age range who enjoy watching movies and TV shows and who want recommendations.

4. Project Concept

This project tackles a subproblem of a larger issue. The larger problem faced by users is effective media-finding: users want to get recommendations for film and television titles from algorithms, determine whether they want to watch it, and then access the shows.

Among other inefficiencies, the recommendation process is opaque. Users cannot understand, visualize, or control why a website’s recommendation algorithm is recommending a particular show, even if that would help them find the show they want. While other problems in the media-finding process are handled by other websites, like Netflix, IMDB, and Goodreads, there is no available source which allows users to clarify and control the recommendation process. Focusing on this subproblem lets our project serve an original and currently-unmet user need.

The design solution is fundamentally to create “links,” or connections, between different film and television titles. Our site will use these connections as a source of recommendations. That means that media titles will be recommended on the basis of criteria like ‘same author,’ ‘same director,’ ‘similar story,’ and so on. By recommending movies and TV shows on the basis of specific, named connections, we allow users to understand why they are being recommended a particular title. Users will also be able to choose which links they want to prioritize. This will allow them to customize why they are receiving particular recommendations.

5. Challenge

The project has a broad scope, and is a significant creative challenge. The problems users face—understanding and controlling their movie and television recommendations—is extremely broad. In order to give users the ability to control their own recommendation process, our design solution has to acknowledge many different priorities from users of all different kinds. It also has to synthesize various different aspects of a movie or TV show in order to find similarities between different titles which exist for different reasons.

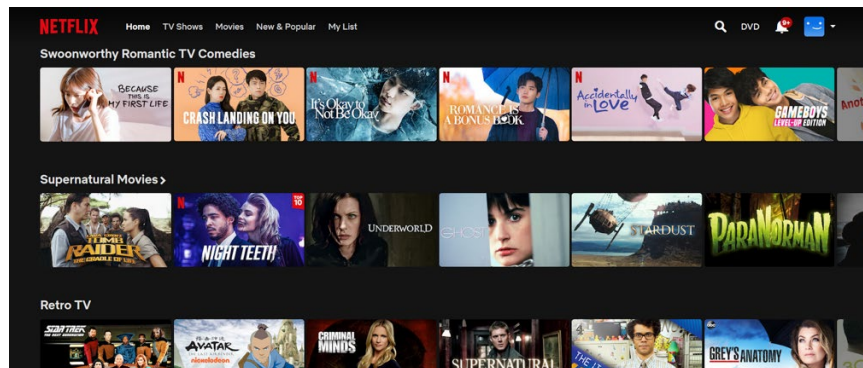
The project is also a creative design challenge. We are reimagining the traditional model of movie and TV recommendations in a way which must be innovative enough to engage users, intuitive enough to be usable, and which will give users understanding and control of the recommendation process. Figuring out how to let users best control their recommendations is a significant challenge.

6. Comparative Analysis

Many popular streaming services (Netflix, Hulu, Disney+, Peacock) use what we call the ‘Blockbusters’ model for recommendations. They show a horizontal array of movie and TV shows covers, similar to a shelf at the defunct video rental service Blockbuster, and each row is recommended based on a criterion chosen by the site, not the user.

These services do well by combining their central service—providing users with movies and TV that they can watch online—with recommendations. This creates a hub where users can find a show and immediately start watching it. They also do well by making the design similar to real-world designs—like shelves at a video store.

However, where they fail is in their lack of explanation and control. Users cannot choose how movies and TV shows are recommended to them, and they do not understand why these titles are recommended. Also, there is no way to get recommendations between streaming services—each service recommends only the titles that it owns, even if there are titles which the user would like better.



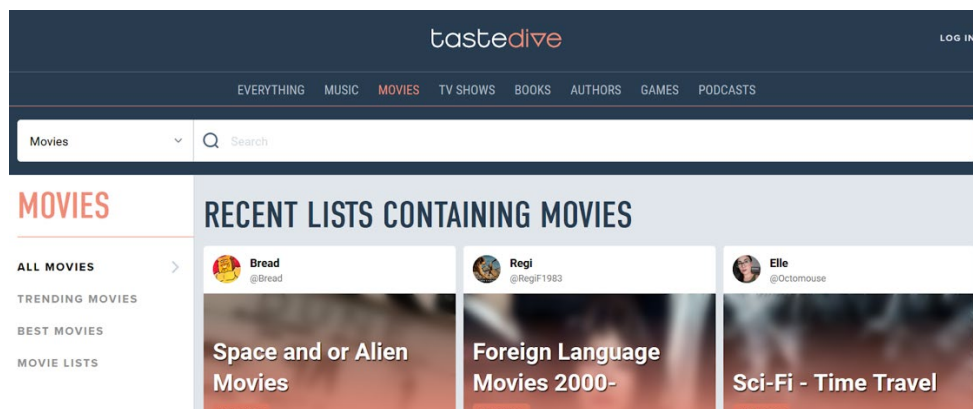
A page from the popular streaming service Netflix showing horizontal rows of visual media. The rows are labeled ‘swoon worthy romantic TV comedies,’ ‘supernatural movies,’ and ‘retro TV.’ There is no way to choose what options are shown.

There are also less-popular sites which are dedicated to finding new movies and TV shows, like TasteDive. TasteDive shows lists curated by individuals collecting different movies, TV shows, books, authors, games, podcasts, or music. It also shows which media titles of each type are ‘trending,’ and which are the ‘best.’ Clicking on an individual media title provides you with a

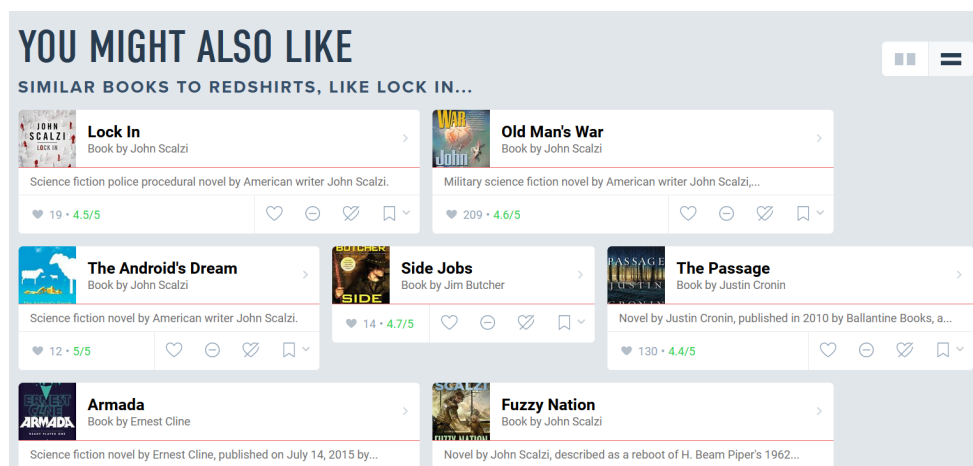
brief summary of the story, links to streaming services where you can watch the title, and a number of media titles which are “similar” to it. No explanation is provided about how this similarity is determined.

TasteDive succeeds by collecting many different types of media in one place. It allows users to see media titles from a number of streaming services, not just one, and gives users access to them (by showing where users can stream that title). It also allows users to create lists and view the lists of other users. These are important efficiencies.

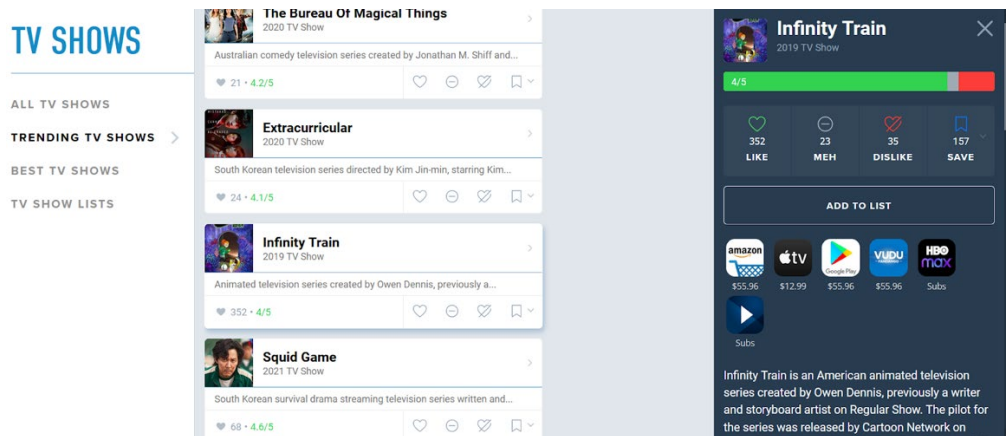
However, like the streaming services, TasteDive does not offer users full control or understanding of their recommendations. The only actual recommendations on the site are based on the undefined characteristics of “trending,” “best,” or “similar,” or through the individually-curated lists.



TasteDive’s front page takes you to different types of media: ‘everything,’ ‘music,’ ‘movies,’ ‘TV shows,’ ‘books,’ ‘authors,’ ‘games,’ and ‘podcasts.’ On the ‘Movies’ page, we see user-curated lists containing movies, trending movies, and ‘best’ movies.



Individual pages for a book show you books which are “similar” without explaining why.



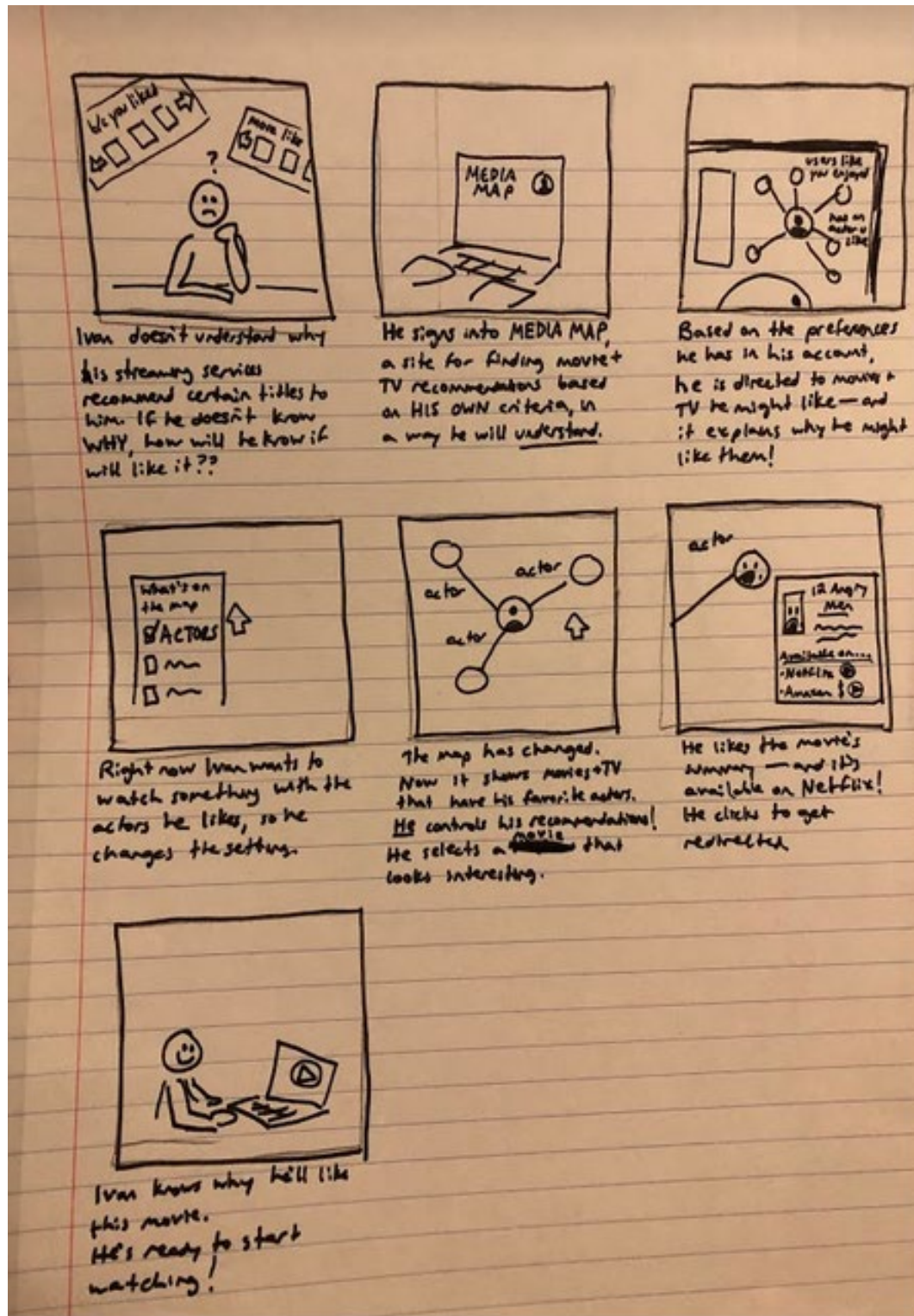
TasteDive provides a summary of the media title, the ability to add it to a list, as well as links to where it is available to stream (and for how much money.)

7. Explanation of design solution

As explained in our project concept, our design solution is to provide more control on the user's end in terms of personalizing the recommendation algorithm of movies and TV shows. In this regard, our design solution is to create a platform that grants users freedom to choose/provide their “key attribute” that is applied towards our recommendation algorithm. This personalized recommendation algorithm will be generated by Relational database or GraphQL database, and we will implement this algorithm to our web application.

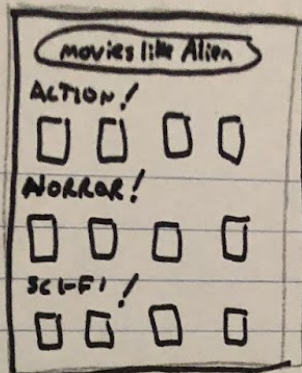
(Updated in Progress Report 2) To make our design more user-friendly, we changed the visual representation of our design solution from the bubble-chart to the grid-chart.

8. Storyboards

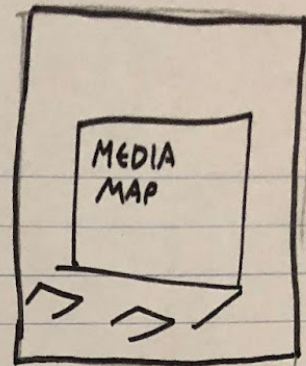




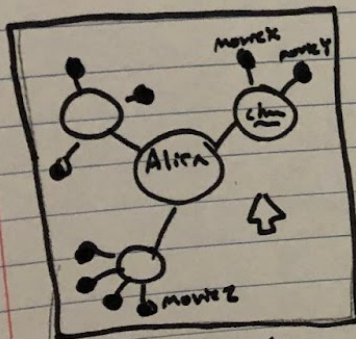
Nina is a big fan of the movie *Alien*, but only the characters — she normally hates that genre.



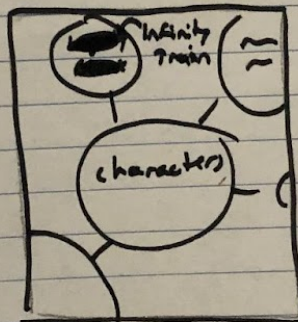
But traditional searches only give her movies + TV similar to *Alien*'s genre — not the characters! She can't control the recommendation to look for what she wants.



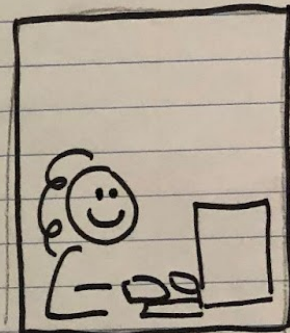
So she goes to Media Map, a site for controlling your recommendations.



Media Map shows her a map of other movies + TV shows with similar attributes to *Alien*.



Look! Here are some movies + shows with similar characters, which aren't necessarily in the same genre!



Nina is happy. Thanks to Media Map, she got to a TV show that she wanted.

9. Current Prototypes

When you first open Media Map (without being signed in), the home page will look like *Fig. 1*.

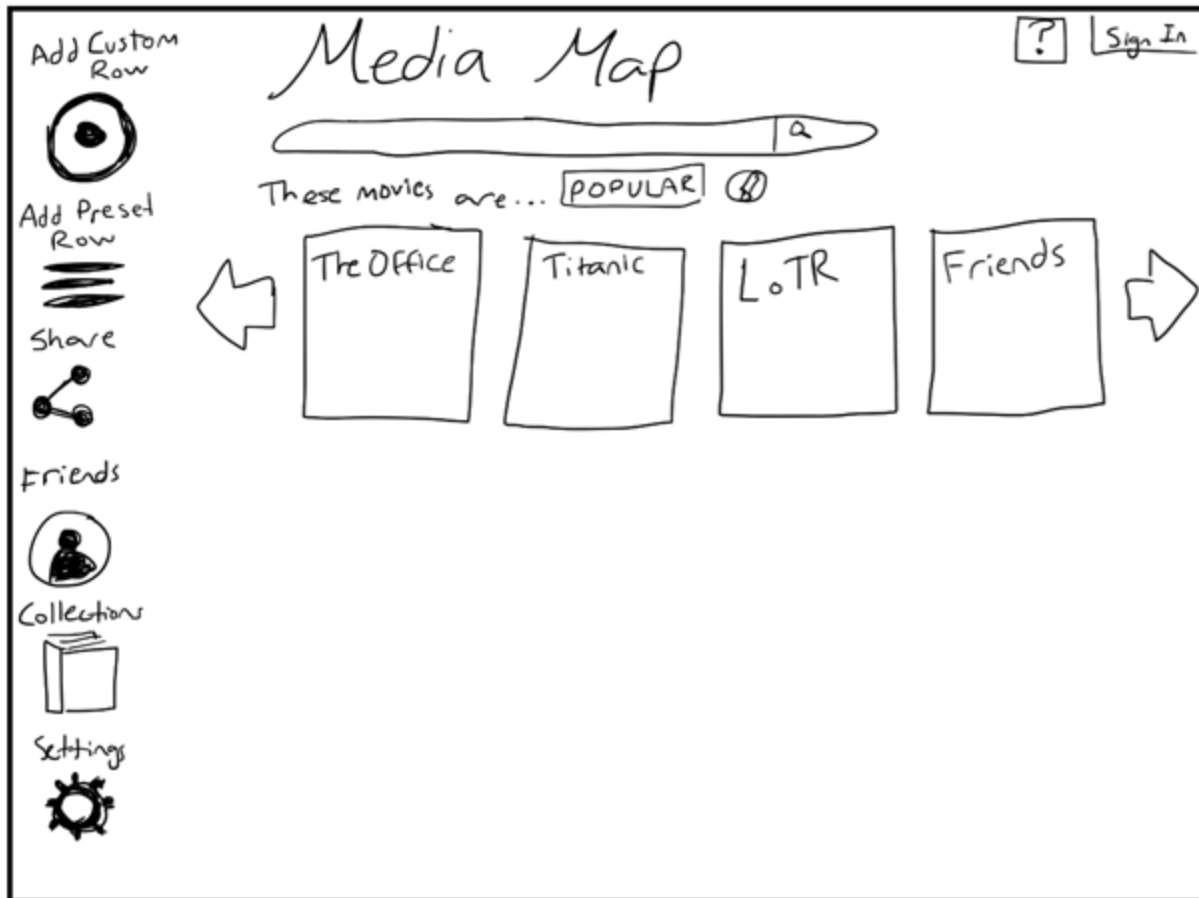


Fig. 1: Home Page

The most prominent element of the page is the website's title – MEDIA MAP – followed by the search bar directly below it.

Below the search bar is the first *row*. The row consists of a label ("These movies are...") and a list of criteria ("popular"), below which is a horizontal array of movies and TV show covers. In this case, the row only has one criterion, 'popular,' so the movies and TV shows in the row are included because they are popular.

The row is here by default. It is intended to show users what it looks like when rows are added to Media Map.

On the left side, there is a sidebar with various options: 'add custom row,' 'add preset row,' 'share,' 'friends,' 'collections,' and 'settings.'

First, let's click on 'add preset row.'

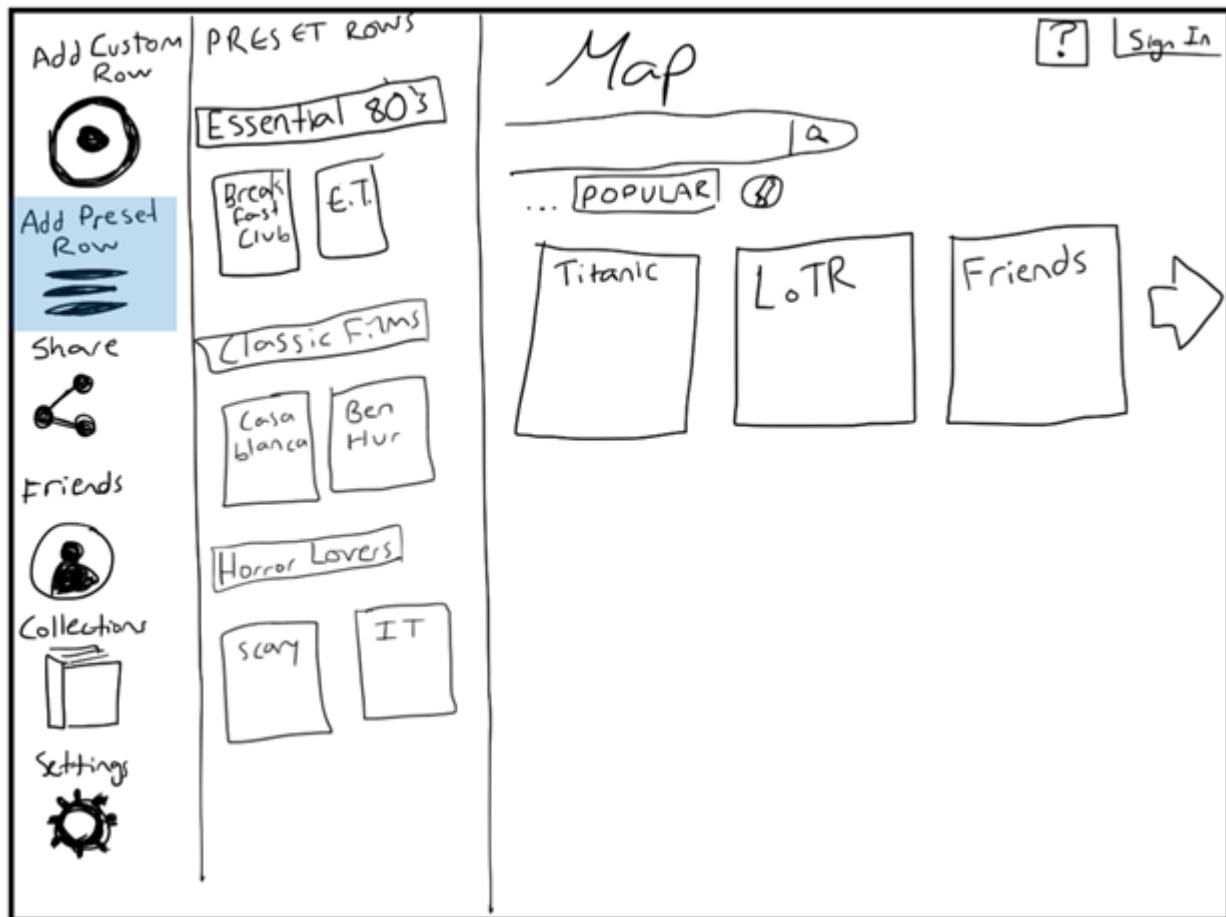


Fig. 2: After clicking “Add Preset Row”

After clicking “add preset rows,” another column appears on the sidebar. In this column, there are a list of *preset rows*, including ‘Essential 80’s,’ ‘Classic Films,’ and ‘Horror Lovers.’

There is a blue rectangle around “add preset rows” simply to show that it has been clicked.

Let's click 'Classic Films.'

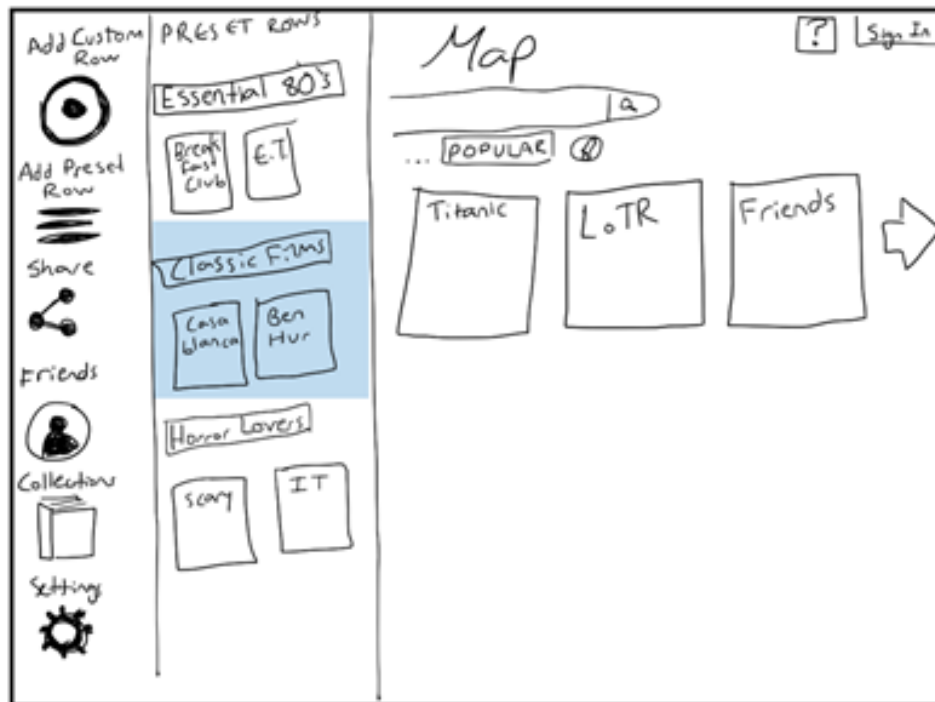


Fig. 4: The screen in the process of clicking "Classic Films"

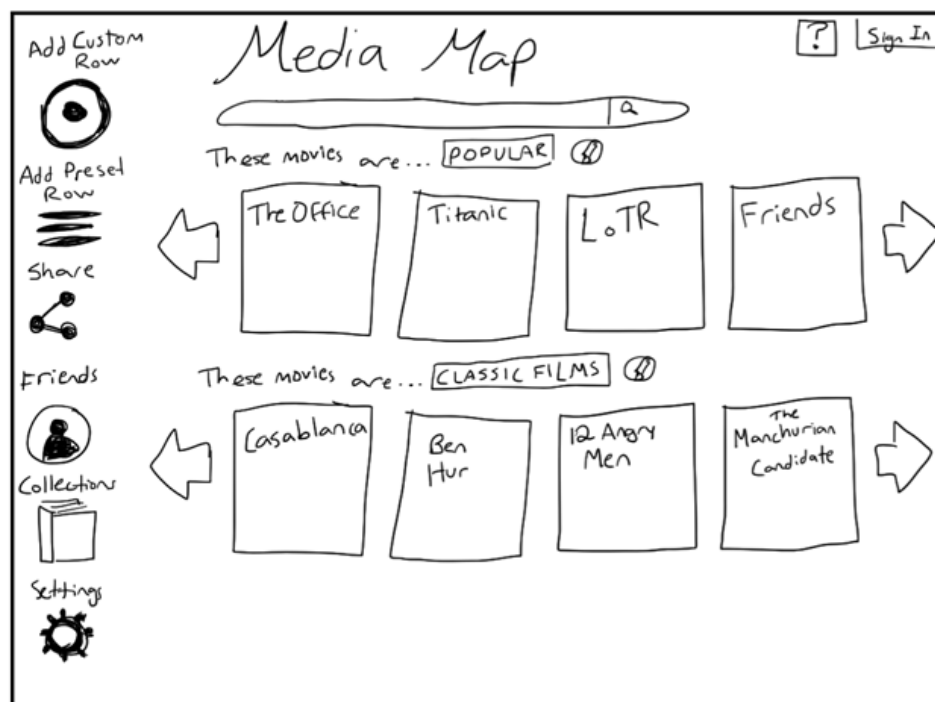


Fig. 5: Result of clicking "Classic Films"

Clicking "classic films" has added a new row to the home page. This row has one criterion, which is "classic films": the name of the preset row.

In order to add a row based on your own criteria, click “add custom row.” This will cause two columns to be added to the sidebar.

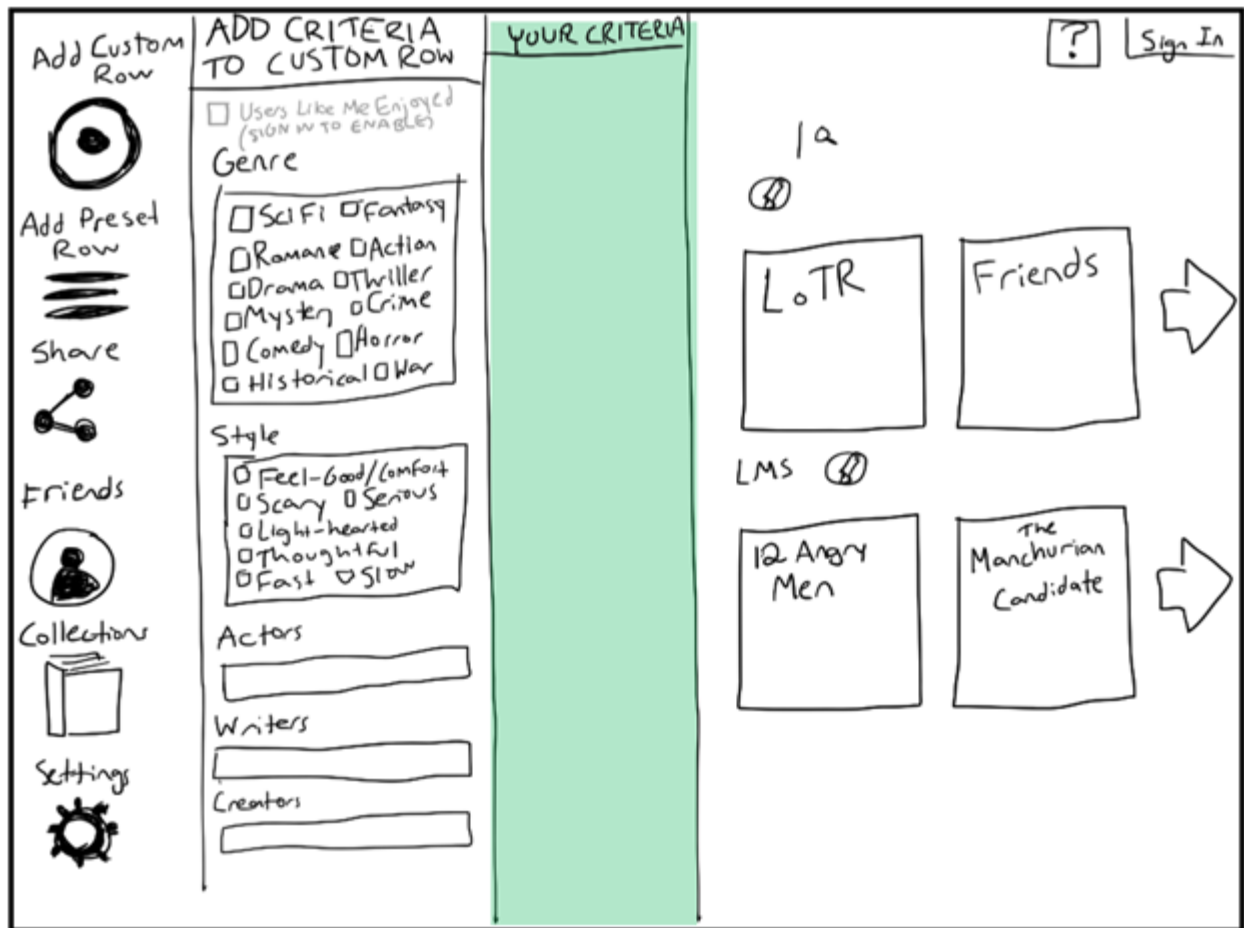


Fig. 6: After clicking “Add Custom Row”

The first additional sidebar column allows the user to choose which criteria which to add to the row. These criteria include genre, style, actors, writers, and other creators. If the user had signed in, “users like me enjoyed” would be another possible criterion. If the “users like me enjoyed” criterion were selected, the row would use collaborative filtering to prioritize the movies and TV shows that the user would most likely enjoy.

The second column shows the user all the criteria that the user has already selected for this row. The “ADD ROW” button at the bottom will create a row based on the criteria that are currently selected.

Let’s try adding a lot of criteria, then hitting “add row.”

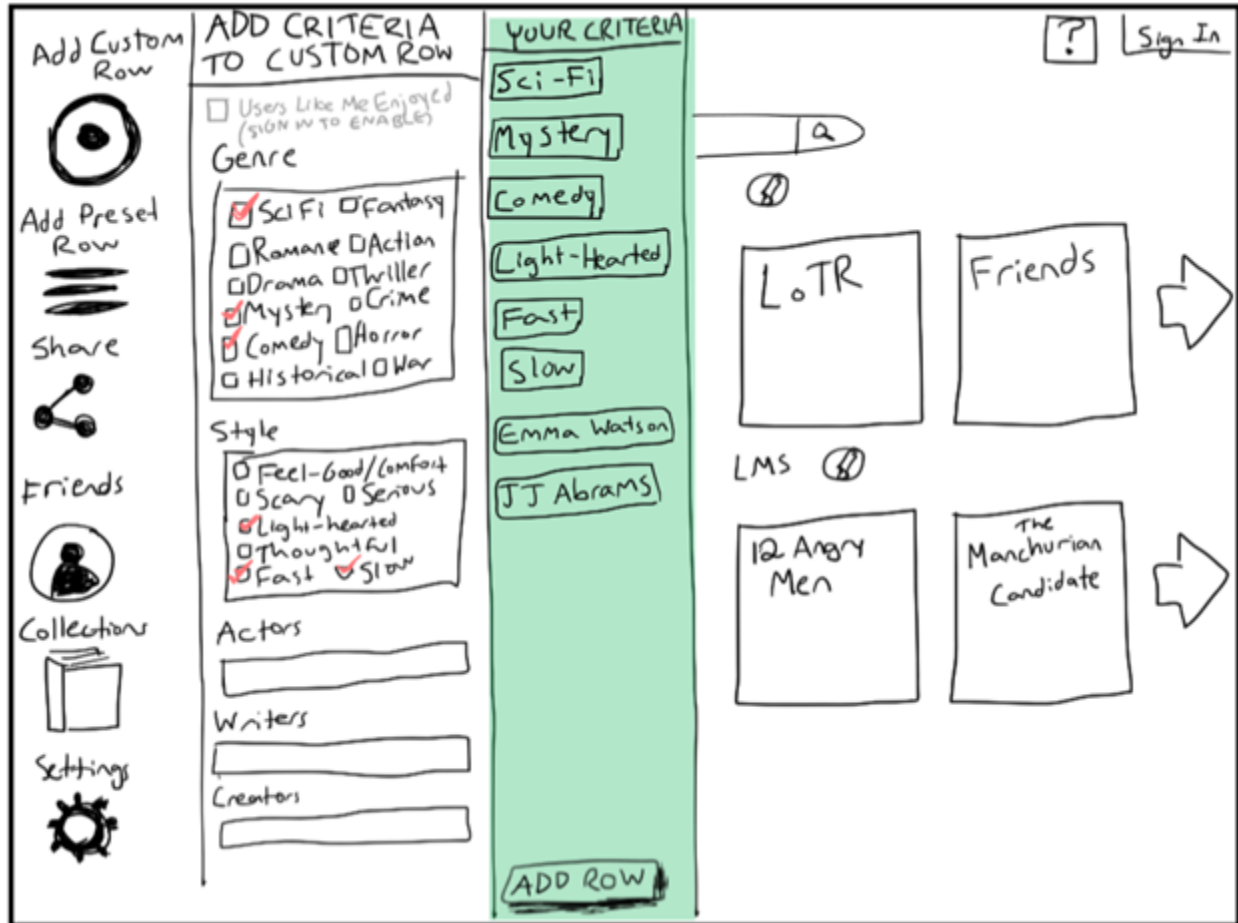


Fig 7: After adding a lot of criteria

After adding this new row...

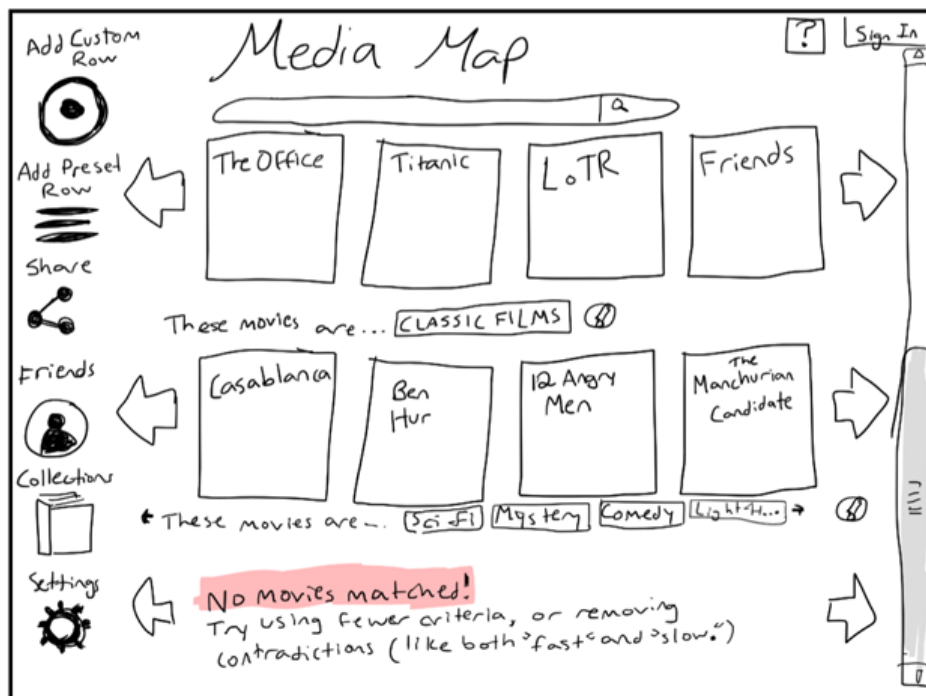


Fig 8: After adding the new custom row

Uh oh! No movies matched, because there were too many criteria, and some were contradictory.

In this case, we can press the edit icon to the right of the newly-added row's label.

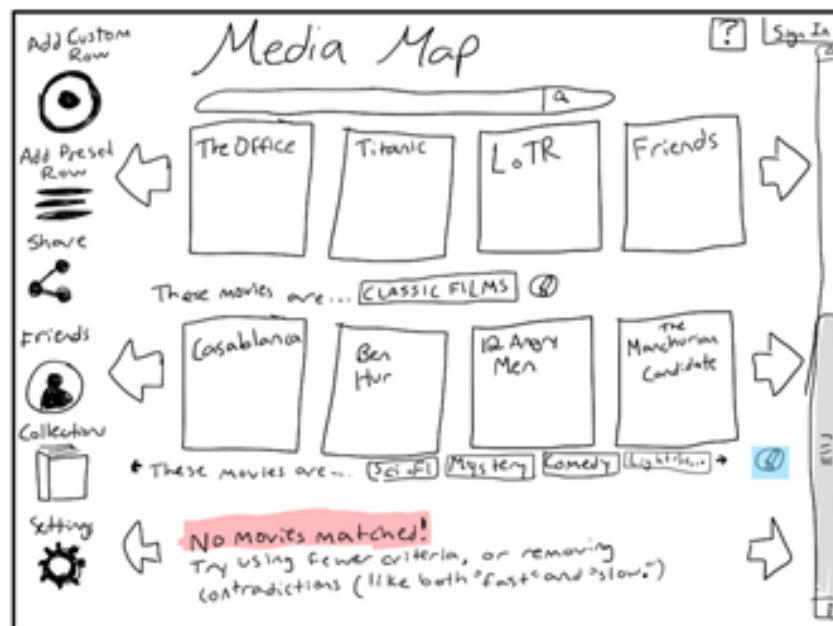


Fig 9: Clicking on the edit icon next to the newly-added row's label

This will reopen the “add custom row” sidebar. From here, you can edit the criteria in the row.

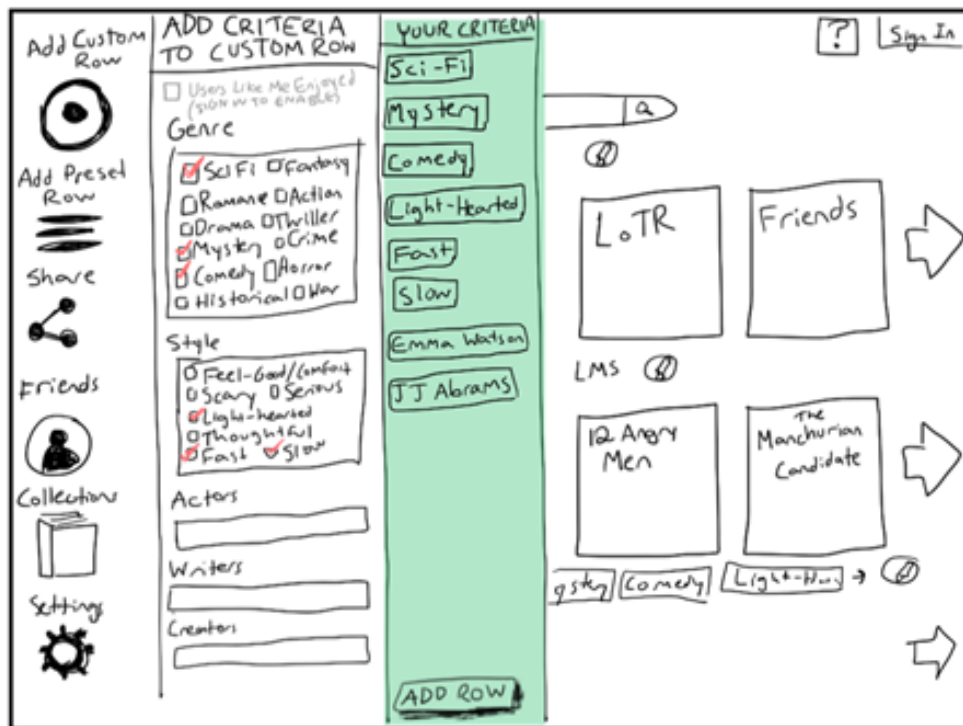


Fig. 10: Reopened “add custom row” in order to edit the criteria.

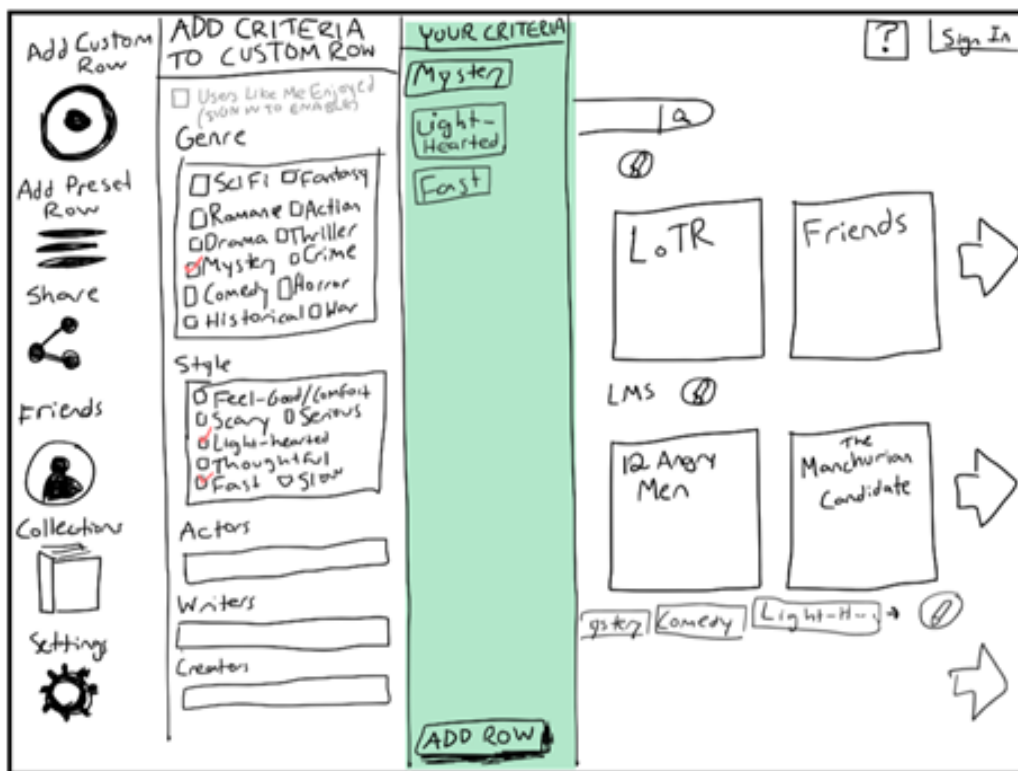


Fig. 11: Edited the criteria in the row

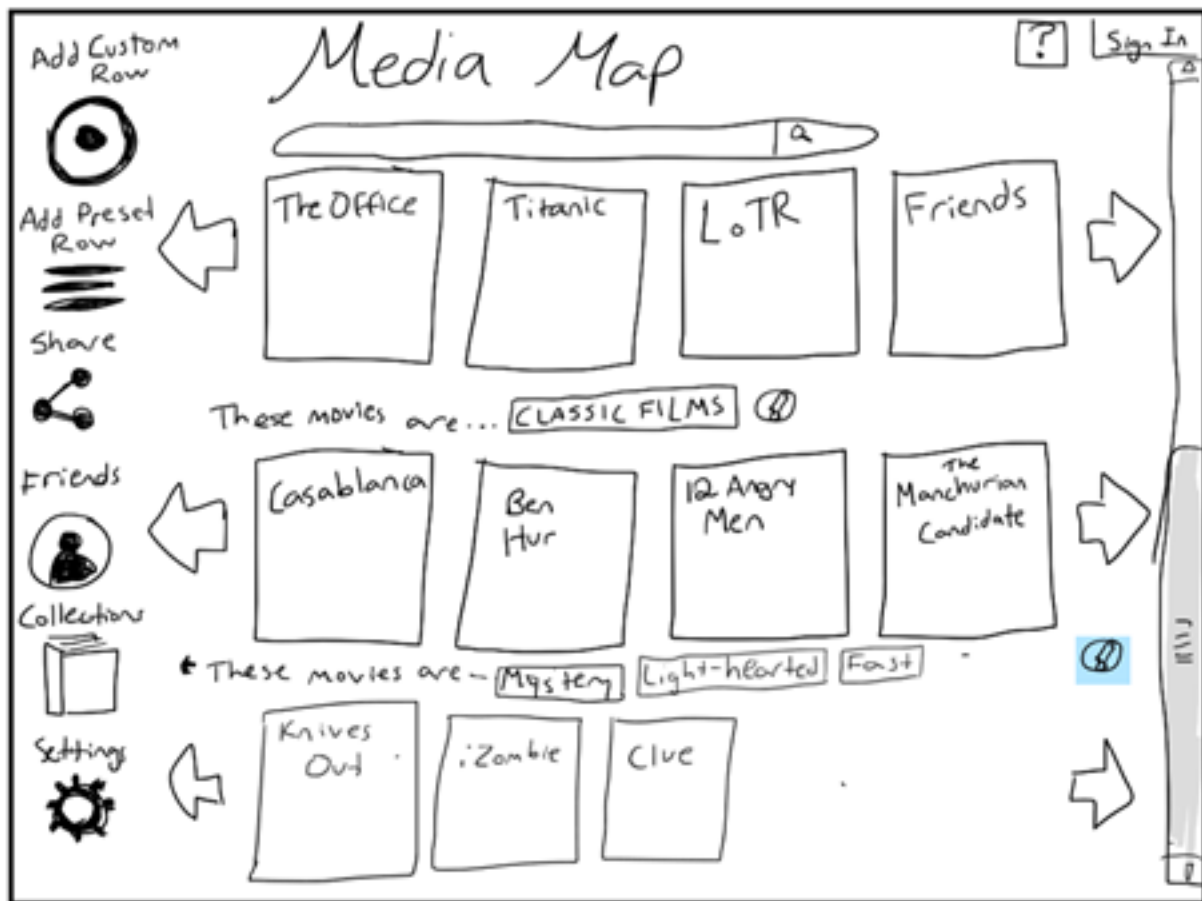


Fig. 12: Updated the row and added it.

Now that we have these new movies and TV shows to look at, let's dig deeper and click on *Clue*.

This takes us to the *Clue* page. The *Clue* page can also be reached by searching for "Clue" in the search bar.

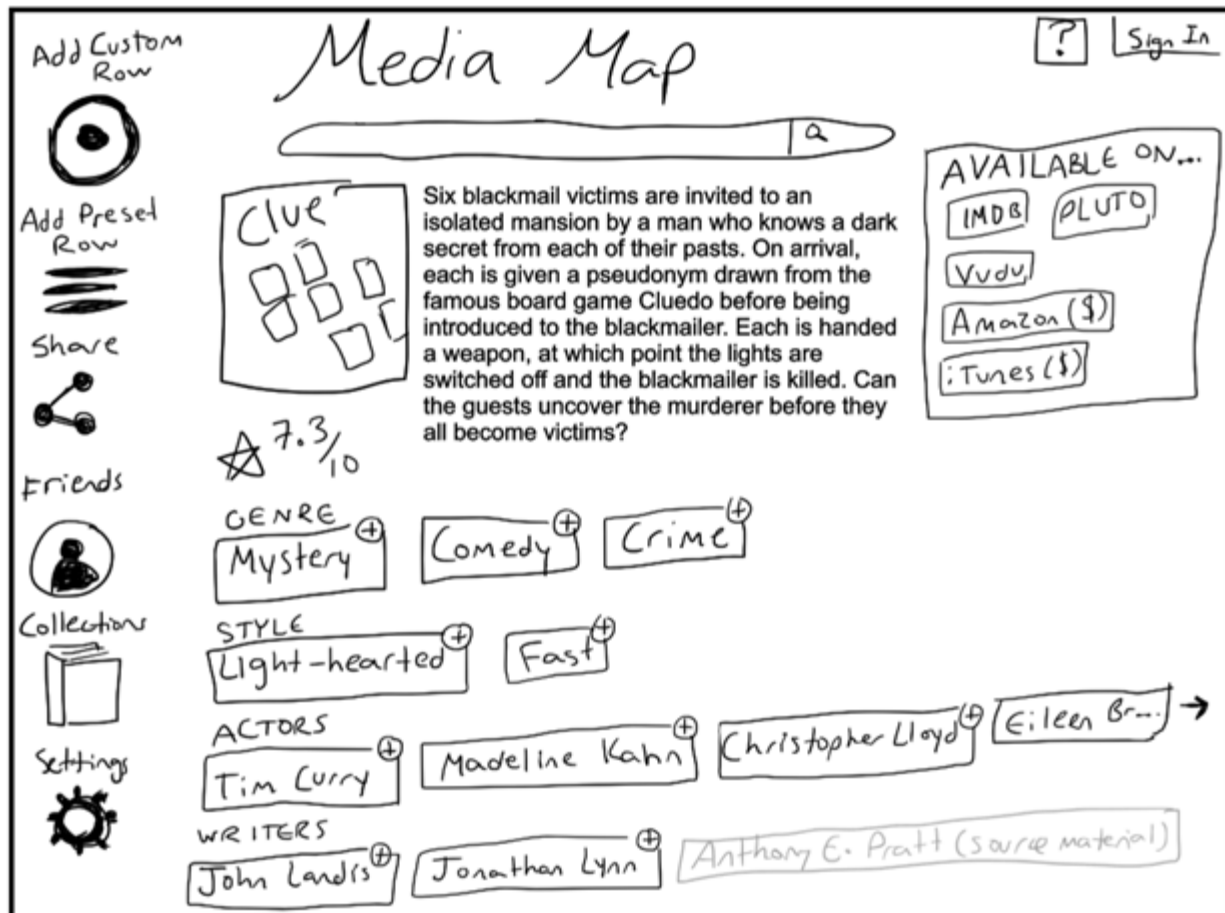


Fig. 12: After clicking on the movie "Clue"

On the page for *Clue*, the most prominent areas (aside from the title of the website and the search bar) are the movie's cover, the movie's summary, and the list of streaming services on which the movie is available. All of these items are large and in the top row. Below the movie's cover is a star rating of the movie, sourced from IMDB.

Below all of this information is a list of the criteria that *Clue* fits. For the criterion of "genre," *Clue* fits mystery, comedy, and crime. For the criterion of "style," *Clue* fits light-hearted and fast.

When a particular criterion does not apply to any other movie or TV show in the Media Map database, that criterion is greyed-out, like for the writer Anthony E. Pratt, who wrote the board game Cluedo.

Every criterion is within a square, and has a plus button in the upper right corner. If the plus button is clicked, that criterion is added to "Add Custom Row." This means that if a user likes a particular movie, like *Clue*, and wants to find other movies that share similar criteria with *Clue*, our website can help them. The user can create a new row by adding different criteria which also apply to the movie *Clue*, allowing the user to find similar movies *while knowing what makes those movies similar.*

Let's go back to the main page.

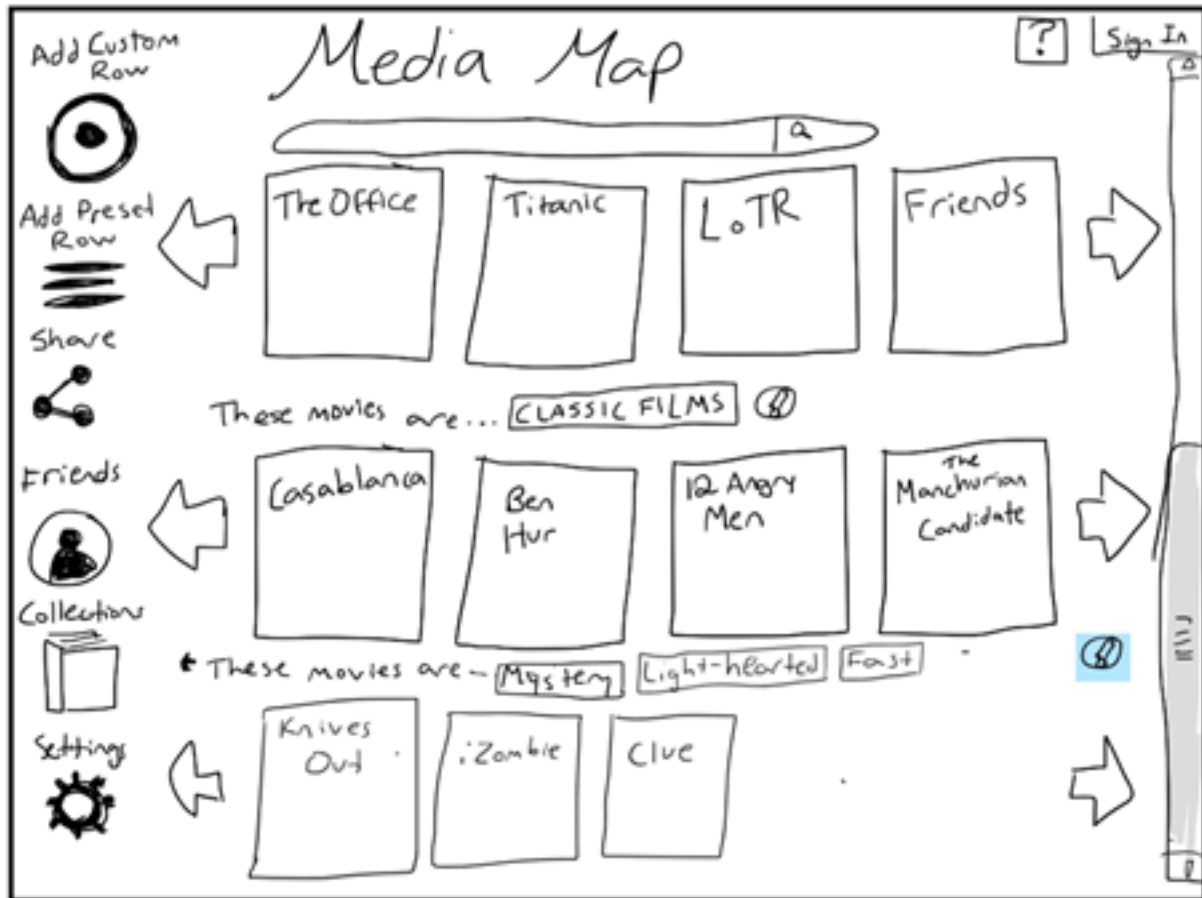


Fig. 13: Back to the main page.

On the upper right-hand corner of the main page, we have

- **SIGN IN:**

This button would allow users to sign in. Users who are signed in would be able to access more functionalities of the site, including recommendations based on their watch histories, access to their friends, etc.

- **?:**

This is the help button. It would show an explanation of how the site works—possibly an explanation similar to this one.

Also on the main page's sidebar, we have:

- **Share/Export Button:**

Share/Export Button will allow the users to share the rows of their current map not only on social media but also with other users of Media Map.

It will also allow users to export the current media map in .mm9 format to store the map locally.

- **Friends:**

If you, the user, are signed in, you can see other users of media map who are connected to you as friends under the “Friends” tab. You can see any saved rows of your friends which are public, and add them to your main page as if they were preset rows. You can also see your friends’ collections, and save them to your own ‘collections’ tab.

- **Collections:**

Collections are like to-watch lists. The user can save specific movies and TV shows to a collection, forming groups of media titles. You can add and remove movies and TV shows from your own collections. If you’ve saved the public collections of your friends, you can also view them in this tab.

The symbol of ‘collections’ should be a bookmark symbol, not a book.

- **Settings:**

Miscellaneous settings (TBA) mostly for display settings will be under setting.

10. Risks to mitigate (Updated in Final Progress Report 2)

(Initial Risks in Progress Report 1) From the Individual Project, we have reason to believe that a user need for our project exists, at least within the confined population of Columbia University students, and likely outside that niche as well. As such, our central risk is finding the best way to answer that need.

The first risk we wish to mitigate is that of the project’s format. Our project should allow users to visualize, understand, and control connections between different movies and TV shows, but there are many ways of doing this—some better than others. By creating numerous designs and showing them to users, we will climb different design hills until we find the format that is best for users: the “right design”.

Other design risks include those of usability. Once we have a format, how do we best utilize that format to make a finished project that is simple, elegant, and easy to use?

On the technical end, our risks are functional: will we be able to meet the standards set by the design? That is, generating the media map with simple and clear choices, such as “same actors” or “same director” is definitely feasible to implement. However, when it comes to the case that if users choose some abstract criteria, such as “similar stories,” to generate the media map it could be harder to provide better recommendations than existing recommendation services like Netflix. On top of that, after generating a “media map” based on the user’s choice of key attributes, we would like to show users which streaming services they can utilize to watch each movie and tv show on the map. However, as the streaming services update the list of movies and tv shows that are available based on regions, contracts, and so on, we might not be able to reflect these updates accurately, which might end up providing outdated information.

(Updated in Progress Report 2) As a result of our second loop, we have decided to modify our final design to the grid style instead of the bubble-chart style, based on the user's preference on its simplicity and better usability.

Now, we need to mitigate two main risks: first, is our renovated design optimal enough for our users? Second, are we able to technically implement the high-standard recommendation algorithm with the list of movies, as mentioned above in our initial risks. Therefore, from now on, we will focus on the programming task to create this design properly into an actual product and the second round of user interviews to make sure our design serves best for our target user population

11. Project Tasks (Updated in Final Progress Report 2)

A list of tasks that must be completed in order to finish the final project. This should include tasks that have been completed so far and tasks that are yet to be completed. For each incomplete task, state who is currently assigned to complete it (assigning multiple people is OK), how many hours you estimate the task will take to complete, and what your internal due date for completion is. For each complete task, state who completed it (stating multiple people is OK), how many hours it took to complete it, and the date that it was completed. Include ALL known tasks here, including implementation (coding), user testing, and project write up tasks.

- TASKS COMPLETED

- Pick design topic (2 hrs, all, 11/13)
- Balsamiq prototypes for different formats (2 hrs, Adi, 11/15)
- Further developed ideas for different formats (1 hr, Adi and Swan, 11/16)
- Conduct research on which Web Frameworks / DBMS we will use for the project (2 hrs, Je Seung and Alexsandra, 11/16)
- FP Progress Report I (all) 11/19
- Create low-fi prototypes for different formats (2 hrs, Adi + Swan). Internal due date: 11/19
- First studio session (all) 11/23
- User interviews regarding different formats (all) 11/29
- New design based on user interviews (all) 12/01
- Lo-Fi prototype the new design (1hr, Adi, 12/02)
- Coding planning & Research on technical stacks (3 hrs, Alexsandra and Je Seung, 12/02)
- FP Progress Report II and 360-assessment (all) 12/03

- TASKS INCOMPLETE

Project Write-up tasks

- final project turn-in (12/16, 11AM)
- final project video (12/15, 3PM)

Design tasks

- User interviews regarding new design (30 min apiece, all) Internal due date: 12/7
- Update design based on secondary user interviews (1 hr) Internal due date: 12/10
- High-fi prototype the updated design (2 hrs) Internal due date: 12/10

Coding tasks

- Layout basic elements (forms, inputs, etc), including sign in and sign up page, profile pages, etc (10 hrs) Internal due date: 12/6 (consider extra work if the design changes)

- Complete the front-end programming and test to finalize the product (20 hrs) Internal due date: 12/13

Coding tasks (not our priority based on Professor Smith's feedback)

- Create Sign in / Sign up APIs - integrate it with front end (5 hrs) Internal due date: 12/7
- Integrate Neo4j API with front end to make element interactive (5 hrs) Internal due date: 12/10
- Implement Neo4j with Flask to create a movie recommendation feature (10 hrs) Internal due date: 12/10
- Integrate both ends and test app (3 hrs) Internal due date: 12/13

- TASK ASSIGNMENT (finalized)

Design: Adi and Swan

Coding: Alexsandra and Je Seung

Part 2: Additional sections for Progress Report 2

Q1: Design Process

- **First Loop:** as our Final Project is an extension of Adi's Individual Project, I am referring the relevant information for this step from our Progress Report 1.

This project tackles a subproblem of a larger issue. The larger problem faced by users is effective media-finding: users want to get recommendations for film and television titles from algorithms, determine whether they want to watch it, and then access the shows.

Among other inefficiencies, the recommendation process is opaque. Users cannot understand, visualize, or control why a website's recommendation algorithm is recommending a particular show, even if that would help them find the show they want. While other problems in the media-finding process are handled by other websites, like Netflix, IMDB, and Goodreads, there is no available source which allows users to clarify and control the recommendation process. Focusing on this subproblem lets our project serve an original and currently-unmet user need.

We came up with the design solution is fundamentally to create "links," or connections, between different film and television titles. Our site will use these connections as a source of recommendations. That means that media titles will be recommended on the basis of criteria like 'same author,' 'same director,' 'similar story,' and so on. By recommending movies and TV shows on the basis of specific, named connections, we allow users to understand why they are being recommended a particular title. Users will also be able to choose which links they want to prioritize. This will allow them to customize why they are receiving particular recommendations.

- **Second Loop**
- **Biggest Risk**

The two biggest risks we aimed to tackle with the UCD loop are like below:

1. To find the most optimal format for this project, as we mentioned in the progress report 1. Our project should allow users to visualize, understand, and control connections between different movies and TV shows, but there are many ways of doing this—some better than others. By creating numerous designs and showing them to users, we will climb different design hills until we find the format that is best for users: the “right design”.
2. Once we have a format, how do we best utilize that format to make a finished project that is simple, elegant, and easy to use?

- **Design/Prototyping Phase**

In the second loop, extending from the first loop of the design process, we revisited the first step of the cycle, which is the design step. Based on the result of team meetings and the feedback we received on the Progress Report 1, we focused on re-evaluating our Need-finding and Storyboarding stage at first. As a result, we all agreed that there is no need to launch completely new need-finding or storyboarding steps. This is because our first user interviews conducted in the first design loop demonstrated that our intended target population, people who are interested in movies or TV shows, has the needs of our “Media Map” as they can understand the logic why they are being recommended some titles.

In that regard, we then revisited the second step of the cycle, which is the prototype step. We decided to come up with five different Lo-Fi prototypes, to conduct user studies on the format of the design, which is this loop’s biggest design risk. So, we have created five Lo-Fi prototypes in total, which include three Lo-Fi bubble-chart-style designs and two grid-style designs, with balsamiq.

Based on Professor Smith’s feedback, we made the designs more individualized. Instead of having generic “movie name here” icons, we added the actual titles of movies and TV shows, connected by actual links reflecting the content of those shows. Further, when users provided us with the titles of shows that they liked, we altered the designs further so that the design appeared personalized to the user’s tastes.

- **Evaluation Phase**

After finishing those designs, we moved on to the third step of the design cycle, the evaluation step. As mentioned above, we have conducted user interviews (sixteen users in total) with the questions we created during our team meeting.

The list and structure of questions are like below:

- PRE-INTERVIEW
 1. How do you usually find movies and TV shows?
 2. Can you tell me about the last X movies and TV shows you’ve watched? How did you find them?
 3. What kind of things do you look for in a new movie or TV show?

Show the design 1

Show the design 2

Show the design 3

Show the design 4

Show the design 5

- POST-INTERVIEW

1. What design was your favorite? Why? What did you like about it? What didn't you?
2. What would your ideal design be able to do?
3. If you could pick and choose different parts of different designs, what would you like?
4. What kind of "criteria" would you want to see on a website like this?

We constructed the interview structure like this, which is not to ask questions about each design before moving on to the next design, as we did not want to interrupt user's focus on exploring each design.

We switched the order of the designs we were showing, so that some users saw bubble-style formats like Designs 1, 2, and 5 first, and other users saw grid-style formats like Designs 3 and 4 first. Due to constraints in synchronizing our work (since we all did interviews separately), as well as the constrained number of interviews done, the different orders of the designs were not truly randomly distributed, which is a potential issue in our research.

As a result of our interviews, most of our interviewees preferred the grid style instead of the bubble-chart style, as it is more simple and easier to get the necessary information, which means the grid style appears to have a higher usability than the bubble-chart one. Therefore, we decided to move on with the renovated grid-style design.

In detail, we decided to re-organize our side bar and top bars since many of our interviewees expressed that they prefer more organized filtering and search system. Also, we decided to make the design more minimalistic as many interviewees preferred that. On top of that, we have added the feature of share/export and friends/collections features. This is because some interviewees expressed their interest in sharing their media map filters to their friends as they do with Spotify (sending their Spotify playlist to their friends).

As a result of this loop, we have updated our prototype, which is listed on Q9 in Part 1 of this progress report (Current Prototype). Also, we have uploaded our updated prototype as our supplementary file.

Based on our updated prototype, our programming team has planned and evaluated what coding tasks we need to work on to finish this project on time.

Q2: Progress Report 1 feedback follow-up (provided by Prof. Smith)

- **In our first meeting after Progress Report 1 feedback**

At our first final project studio, Professor Smith emphasized that the design portion of our project is more important than the technical implementation. In this regard, in our first post-studio team meeting, we decided to conduct user interviews with our five different balsamiq prototypes, to fine-tune what design components we need to implement to make our project more user-friendly and useful. In that sense, we created interview questions and conducted sixteen user interviews over the week.

- **In our second meeting after Progress Report 1 feedback**

In this meeting, we evaluated the result of user interviews, and then modified our design based on the user's response and Professor Smith. That is, since Professor Smith and the most of interviewees answered that the grid-style design is more usable due to its simplicity, we came up with the modified lo-fi design in the grid-style during the meeting. Then, we also modified smaller details on our final lo-fi design based on the user interview, to mitigate our design risks by finding the most optimal design.